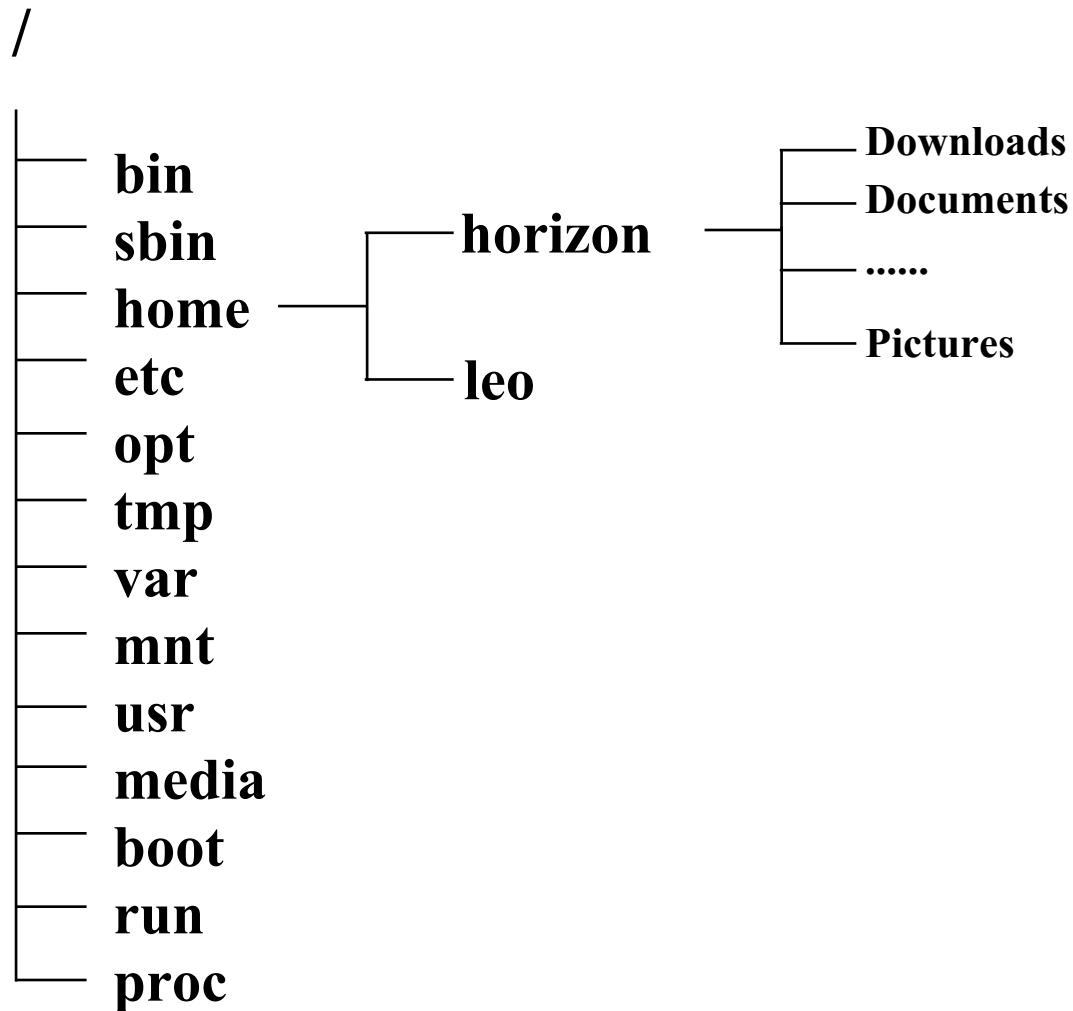


File Structure of Linux



/ :

‘/’ Is the root directory or the starting point of linux file system

/home:

The home directory contains the folder for each user for e.g. /home/horizon

/bin: (binaries)

Contains user binaries (commands, programs).

E.g. Cp, mkdir, pwd, ls, etc.

/sbin: (system binaries)

Contains system administration binaries

E.g. Fdisk, useradd, userdel, etc.

/usr:

Contains read-only commands, libraries and data.

- /bin is linked to /usr/bin.
- /sbin is linked to /usr/sbin.
- /lib is linked to /usr/lib.

/etc:

Contains system-wide config files.

E.g.:

- /etc/fstab
- /etc/sysconfig/network-script

/opt:

Opt stands for optional.

Used by 3rd party softwares

/media:

Mounts temporary media device (USB, Hard Drive, etc.)

/boot:

Contains grub, bootloader, kernel, etc.

/mnt:

Temporary mount directory.

Commands

The Command prompt lists:

Username whos logged in (e.g. Horizon)

Server hostname (e.g. horizons-den)

Current directory (e.g. ~/Desktop)

"\$" prompt (if you are normal user) or "#" prompt (if you are root user)

```
horizon@horizons-den:~/Desktop$
```

```
root@horizons-den:/home/horizon/Desktop#
```

- **Basic Commands:**

Command Syntax:

Command [options] [argument]

Command: Name of command

Options: Start with one or two dashes (e.g. -a or --a)

Arguments: A target that the command should operate on '[]' means optional.

Commands:

pwd: Shows the user's present working directory.

clear: Clears the contents of the screen.

mkdir: Create a directory.

Syntax: mkdir [options] foldername

Options: -p: Creates parent directories as needed; ignores existing directories without throwing an error.

-m: Sets the file mode (permissions) directly upon creation (e.g., 755).

-v: Prints a confirmation message for every directory created.

-z: Sets the SELinux security context (advanced).

ls: Lists the content of a directory

Syntax: ls [options] [foldername]

Options: -l: List more details of contents inside directory.

-d: Shows information about directory rather than listing.

- a: Shows all files, including hidden files.
- h: To see human readable size.

du: Tells the Disk Usage of the active or mentioned directory.

Syntax: du [options] [directories]

Options: -h: Prints size in easy-to-read format

-s: Only displays the total size of the specified directory, while not displaying the sub-directories

-a: Displays all the files.

-c: Adds a line showing the total of the directory's size

cd: Used to switch between directories.

Syntax: cd [foldername]

Special Characters:

- "." means current directory.
- ".." means parent directory.
- "-" would take you to previous working directory.
- "~" would take you to home directory of the user.

Absolute path: Starts with root directory (i.e "/") & is complete path.

e.g.: cd /home/horizon/Desktop/

Relative path: It is relative to current location & does not start with root directory (i.e "/").

e.g.: cd /horizon/Desktop/

cat: Used to display contents of a file and also to create one.

- To create a file use:

e.g.: cat > hello.txt (Finish by pressing ctrl+d)

- Displaying a file:

E.g.: cat hello.txt

touch: Functions of touch:

- Updates timestamp of file, if exists.

- Creates a new file, if it does not exist.

E.g.: touch abc.txt

cp: Copies file or directory.

Syntax: cp [source] [destination]

mv: Move or rename the folder.

Syntax: mv [source] [destination]

- Move: mv abc.txt /home/Documents

- Rename: mv abc.txt def.txt

rm: Deletes files or directories

```
horizon@horizons-den:~$ df
Filesystem      1K-blocks    Used Available Use% Mounted on
tmpfs            1606684      2520   1604164   1% /run
/dev/nvme0n1p2 490825920 282645480 183174348 61% /
tmpfs            8033412        0   8033412   0% /dev/shm
tmpfs            5120          8     5112   1% /run/lock
efivarfs         268         213        51  81% /sys/firmware/efi/efivars
tmpfs            8033412        64   8033348   1% /tmp
/dev/nvme0n1p1  306584      7940   298644   3% /boot/efi
tmpfs            1606680      112   1606568   1% /run/user/1000
```

Syntax: `rm [options] fileName/folderName`

Options: `-r` or `-R`: Removes directories and their contents recursively.

`-f`: Ignores nonexistent files and never prompts for confirmation

`-i`: Prompts for confirmation before every single removal.

rmdir: Deletes directory (Only empty ones)

Syntax: `rm directory`

df: Lists total amount of disk usage.

- **Advance Commands:**

whoami: Displays name of the user who's logged in
Syntax: whoami

users: Displays all the username who are currently logged in
Syntax: users

who: Displays users currently logged with more details
Syntax: who [options]
Options: -H, -b

```
horizon@horizons-den:~$ who -H
NAME      LINE      TIME      COMMENT
horizon   tty2      2026-07-25 13:42 (:0)
horizon   pts/0     2026-07-25 13:42 (:0)
horizon   pts/1     2026-07-25 19:54 (:0)
horizon@horizons-den:~$ who -b
system boot 2026-07-25 13:42
```

w: Shows whos logged in and what are they doing
Syntax: w

uname: Displays the name of the OS.
Syntax: uname [options]
Options: -a, -r, -n

```
horizon@horizons-den:~$ uname -a
Linux horizons-den 7.0.0-28-generic #28~24.04.1-Ubuntu SMP PREEMPT_DYNAMIC Wed Jul 1 15:50:57 UTC 2 x86_64 x86_64
x86_64 GNU/Linux
horizon@horizons-den:~$ uname -r
7.0.0-28-generic
horizon@horizons-den:~$ uname -n
horizons-den
horizon@horizons-den:~$
```

uptime: Shows below details:
Current time
System's uptime
How many users are logged in af of the moment
System load avg. For past 1, 5 and 15 minutes

Syntax: uptime

(Note: From RedHat 8 when we use above commands only root is listed no other users are listed)

date: Shows Date and time.
Syntax: date

timedatectl: Control the system time and date.

Syntax: timedatectl [options] [arguments]

ifconfig: Displays the IP of the system.

Syntax: ifconfig

hostname: Displays the fully qualified name of server.

Syntax: hostname

free: Displays amount of free and used memory in the system in bytes.

Syntax: free [options]

-k: Shows values in KBs

-m: Shows values in MBs

-g: Shows values in Gbs

shutdown or poweroff: Shutdown the machine immediately or schedule a shutdown using 24 hour format.

Syntax: shutdown [options] [time in 24 hour format]

Options: -r: Reboot

-h: Halt or Shutdown

-P: Poweroff the system

-H: Halt but only the OS shutdowns system stays on.

-c: Cancel pending shutdown

-k: Kick / Fake out. Doesn't actually shut down the system, but sends the warning messages to all users.

reboot: Reboot systems automatically.

Syntax: reboot

which: Shows the full path of the command.

Syntax: which [command_name]

E.g.:

```
horizon@horizons-den:~$ which htop
/usr/bin/htop
horizon@horizons-den:~$
```

whereis: Locate the binary, source, and manual page files for a command.

Syntax: whereis [command_name]

sleep: Suspends the shell by making it inactive for specified seconds.

Syntax: sleep [seconds]

history: Shows last few commands fired by the current user.

Syntax: history [options]

Options: -c: Clears the history

!: Command for finding the command(from history)

```
499 exit
500 neofetch
501 history
```

```
horizon@horizons-den:~$ !500
neofetch

  `./ossyyyyssso/::
  ::oyyyyyyyyyyyyyyyyyyyo:~
  -oyyyyyyyodMMyyyyyyssyyyyo-
  -syyyyyyyydMMyooyyydMMMyyyys-
  oyyysdMysyyysdMMMMMMMMMMMMMyyyyyyo
  `oyyydMMMMysyysoooooodMMMMMyyyyyyyo`
  oyyyyydMMMMMyyyyyyyyyyyysdMMysssssyyyyo
  -yyyyyydMysyyyyyyyyyyysdMMMMMyssyy-
  oyyyysoodMyyyyyyyyyyyyyyydMMMMMyssyyo
  yyyysdMMMMMyyyyyyyyyyyysosyyyyyy
  yyyysdMMMMMyyyyyyyyyyyysosyyyyyy
  oyyyyysosdyyyyyyyyyyydMMMMMyssyyo
  -yyyyyydMysyyyyyyysdMMMMMyssyy-
  oyyyyydMMMyssyyyyyyysdMMyooyyyoyyo
  `oyyydMMMyssyyoooooodMMMMyooyyyyyyo
  oyyysyoyyyysdMMMMMMMMMMMMMyyyyyyo
  -syyyyyydMMMyssyydMMMyssyyys-
  -oyyyyyydMMMyyyyyysosyyyyo-
  ./oyyyyyyyyyyyyyyyo/.
  `./ossyyyyssso/::`

horizon@horizons-den
-----
OS: Kubuntu 24.04.4 LTS x86_64
Host: Aspire A515-57G V1.24
Kernel: 7.0.0-28-generic
Uptime: 1 hour, 7 mins
Packages: 3367 (dpkg), 24 (flatpak), 24 (snap)
Shell: bash 5.2.21
Resolution: 1920x1080
DE: Plasma 5.27.12
WM: KWin
Theme: [Plasma], Breeze [GTK2/3]
Icons: YAMIS-Cosmic [Plasma], YAMIS-Cosmic [GTK2/3]
Terminal: konsole
Terminal Font: IBM Plex Mono Medm 10
CPU: 12th Gen Intel i5-1240P (16) @ 4.400GHz
GPU: NVIDIA GeForce RTX 2050
GPU: Intel Alder Lake-P GT2 [Iris Xe Graphics]
Memory: 5530MiB / 15690MiB
```

ping: Used to check if a machine is reachable or not on the network.

Syntax: ping [ip-address / hostname]

(It will keep working until ctrl + c isn't pressed)

man: Displays manual pages of commands.

Syntax: man [command]

(Press 'Q' to exit)

whatis: Provides very brief descriptions of the command.

Syntax: whatis [command]

E.g.: whatis shutdown

echo: Displays custom message.
Syntax: echo "[message]"
echo "[message] \$(command)"
echo "[message] 'command'"

alias: Sets an alias (similar to nickname) for a command.
Syntax: alias 'shortcutname=command'
E.g.: alias 'cls=clear'

unalias: Remove an alias.
Syntax : unalias shortcutname
E.g.: unalias cls

There are 3 kinds of shell in RedHatOS:

- sh
- bash
- nologin (Exclusively for system users)

To check the active shell, echo \$0 OR echo \$SHELL

To check the installed shells use cat /etc/shells

- **In-built editors in Linux:**

CLI text editors:

vim, vi, nano

We'll focus on Vim editor.

To create a file:

vim file name

E.g.: vim abc.txt

You can't start to type randomly so to do that first press "i" which enables insert mode.

Press "esc" to turn off insert mode so you can use command.

:w (to save)

:q (to quit)

:wq (to do both)

yy (to Copy the whole line)

p (to paste)

<number>yy (to copy number of lines from cursor)

<number>p (to paste multiple lines)

u (to Undo)

dd (to delete the whole line)

<number> dd (to delete the number of lines)

q! (to close forcefully)

set nu (to set line numbers)

set nonu (to un-set line numbers)

<number>G (to jump to number of line)

G (to jump at the end of the line)

/word (to search the word (press "n" to next))

%s/find/replace (for find and replace)

:match Underlined [/word/] (to underline the word)

:match Underlined [^%number\word/] (\%number\ for line of word

e.g. :match Underlined[^%2\apple/)

:match MakeBold [/word/] (To Bold)

:match MakeBold [^%number\word/]

:match MakeItalic [/word/] (To make word Italic)

:match MakeItalic [^%number\word/]

- **User & group theory**

Users:

To login to any UNIX machine, you need a user.

There are 3 types of users in Linux:

- Superuser or root
- System users or Application users
- Local users

Superuser (root):

Automatically created when you install Linux.

Superuser can execute commands under /usr/sbin & /usr/bin.

System users (Application users):

Created automatically during application or software installation.

E.g.: When the user installs Steam a user named “steam” is created automatically.

Local users (Normal users):

Created by root user.

Local user can execute commands under /usr/bin.

Group:

Group is a collection of users.

Users are grouped together for granting common permissions to all members of a group.

A user can be a member of more than one group.

Types of Group:

- Primary Group
- Secondary Group (Supplementary group)

Primary Group:

Automatically generated while creating a user.

The group name and user name are same here.

One user can only have only one primary group.

Secondary Group (Supplementary Group):

Secondary group are created separately with the help of groupadd command.

Multiple users can be added in the secondary group

One user can have more than one secondary groups.

Files containing user and group details:

User Config files:

Filename: /etc/passwd

Contains user details like username, uid, etc.

Structure of fields in passwd: (the symbol ':' is a separator)

horizon : x : 1000 : 1000 : : horizon : /home/horizon : /bin/bash
username : passwordProtected : userId : groupId : comment : Path of home dir :
Path of shell dir.

UserId ranges:

0 for root

1-200 for system users assigned by OS

201-999 for system users used by system processes

1000+ for local users

Shell Types:

There are 3 types of shell (in RedHat):

- sh
- bash (Default Linux shell)
- nologin (The hidden one)

Filename: /etc/shadow

Contains user's password details (if we open the file we'll get to see that the password is stored in Hashcode)

E.g.

horizon:\$adfaufhawfbawff9af sfesfadi34/aefwaafdaef

Structure of fields in shadow

Horizon : !: 12412 : 0 : 99999 : 3 : : :

username : EncryptedPassword : Password age

(! Means account is protected)

(!! Means account isn't protected)

Group config files:

Filename: /etc/group

Contains group details like groupname, etc.

E.g.: Project : x : 1005 : jack, horizon

GroupName : Password Protection : gID : Group members

Filename: /etc/gshadow

Contains the password information for group accounts.

E.g.: Project : ! : : jack, horizon

GroupName : Password encryption : Users that are group admins :

Group members

User commands:

useradd: Creates a new user.

Syntax: `useradd [options] username`

Options: `-g`: Assign primary group.

`-G`: Assign secondary group

`-u`: Assigns specific UID.

`-s`: Change usershell (e.g. `useradd -s /sbin/nologin john`)

`-d`: Change user's home directory.

On adding a new user, a user have below things set:

- Home directory: `/home/username`
- Default shell: `/bin/bash`
- Primary group: Named same as username
- Password: No password is set

passwd: Assign / change user's password.

Syntax: `Passwd username`

chage: Changing the password aging information for user.

Syntax: `chage [options] username`

Options: `-l`: Show account's aging information.

su: Allows user to switch between different user accounts.

Syntax: `su [-] username`

E.g.: `su - horizon`

(Note: `"-"` give shell access, if `"-"` isn't used then nologin shell is used)

sudo: Allows a user to be a permitted to run a command as root, based on the settings in the `/etc/sudoers` file.

Now to modify the user information we use following commands,

usermod: Modify existing user.

Syntax: `usermod [Options] username`

Options: `-g`: Change user's primary group.

Example: `usermod -g groupName userName`

`-G`: Changes user's secondary group.

Example: `usermod -G secondaryGroupName userName`

`-L`: Locks (i.e. Disables) user

`-U`: Unlocks the user

`-s`: Changes the user's login shell

Example: `usermod -s [shellName] userName`

- d: Changes the home directory
- m: Create user's new home directory
- u: Changes user's name.

Example: `usermod -u [oldUsername] [newUsername]`

(**NOTE**: Deleting the user isn't considered as a good practice, so its better if the user is locked (i.e. disabled).)

userdel: Deletes user and removes its entry from '/etc/passwd' and '/etc/shadow/' files. Although the basic userdel command doesn't deletes the home dir of user.

Syntax: `userdel [options] username`

Options: -r: Removes user alongside its home directory.

-f: Force removal of user its files and also its group.

The /etc/skel Directory:

The /etc/skel i.e. The skeleton directory serves as a default template for creating home directories for new user accounts on Linux system.

/etc/skel has the following files:

- .bash_logout
- .bashrc
- .profile

Now the following files get copied in its home dir whenever a new user is created:

- .bashrc
- .bash_logout

It gets copied as same name.

The .profile file contains users login settings but that doesn't get copied in the newly formed user.

Group Commands

groupadd: Create a new group.
Syntax: groupadd <groupName>

groupdel: Delete a group.
Syntax: groupdel <groupName>

Permissions

File or directory permissions are used to control who is able to read, write or execute.

Example:

drwxr-xr-x 2 horizon horizon 4096 Jul 9 08:24 Music

Permission

File type:

- : Regular file

d: Directory

l: Symbolic link

s: Unix domain socket

p: Named pipe

c: Character device file

b: Block device file

Types of users

(***d*** rwxr-xr-x) d: Type of file

(d ***rw**x* r-xr-x) rwx : Owner's access (Symbol - u)

(drwx ***r**-x* r-x) r-x : Group's access (Symbol - g)

(drwxr-x ***r**-x*) r-x : Other Access (i.e. Users) (Symbol - o)

Types of permissions

r: Read

w: Write

x: Execute

Numerical representation

r: 4

w: 2

x: 1

-: 0